

ECE269: Linear Algebra and Applications
Winter 2026
Quiz 2 Solutions

Instructions:

- This is a **closed-book** quiz. We have provided you with the relevant axioms on the second page. The use of any other resources during the quiz is **strictly prohibited**.
 - You are not allowed to share this quiz with anyone or post it online. If you violate this policy, it will be considered a breach of academic integrity as well as copyright, and will be promptly reported. If Prof. Pal suspects a potential breach of academic integrity, she will hold oral examinations with the students concerned. Violation of academic integrity can lead to academic probation.
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1. (**3 × 6 = 18 marks**) Suppose that $\mathcal{V}$ is a subspace of $\mathbb{R}^n$. Let

$$\mathcal{V}^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}^\top \mathbf{y} = 0, \forall \mathbf{y} \in \mathcal{V}\}$$

be the set of vectors orthogonal to every element in $\mathcal{V}$.

- (a) Suppose that $\mathcal{V} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ for some $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$. Express $\mathcal{V}$ and $\mathcal{V}^\perp$ as subspaces induced by the matrix $\mathbf{A} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k] \in \mathbb{R}^{n \times k}$ and its transpose $\mathbf{A}^\top$.
- (b) Show that $\dim(\mathcal{V}) + \dim(\mathcal{V}^\perp) = n$.
- (c) Show that $\mathcal{V} \subseteq \mathcal{W}$ for another subspace $\mathcal{W}$ implies $\mathcal{W}^\perp \subseteq \mathcal{V}^\perp$.

Solutions: See the solutions for HW. 3 problem 1.

2. (**2 × 6 = 12 marks**) **Matrix Rank Identities** Prove or disprove the following statements:

- (a) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$, then $\text{rank}(2\mathbf{A} - \mathbf{I}) = n$ iff $\mathbf{A}^2 = \mathbf{A}$.

Solution: We provide a counterexample, let $\mathbf{A} = 2\mathbf{I}$, then $2\mathbf{A} - \mathbf{I} = 3\mathbf{A}$, and $\text{rank}(3\mathbf{I}) = n$, but $\mathbf{A}^2 = 4\mathbf{I} \neq \mathbf{A}$, therefore $\text{rank}(2\mathbf{A} - \mathbf{I})$ doesn't imply that $\mathbf{A}^2 = \mathbf{A}$.

The other direction is true though. Let $\mathbf{A}^2 = \mathbf{A}$, then $(2\mathbf{A} - \mathbf{I})(2\mathbf{A} - \mathbf{I}) = 4\mathbf{A}^2 - 2\mathbf{A} - 2\mathbf{A} + \mathbf{I} = 4(\mathbf{A}^2 - \mathbf{A}) + \mathbf{I} = \mathbf{I}$ because $\mathbf{A}^2 = \mathbf{A}$. Therefore, $(2\mathbf{A} - \mathbf{I})$ is invertible and thus has full rank.

(b) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times k}$, then

$$\text{rank}(\mathbf{B}^\top \mathbf{B}) \leq \text{rank}(\mathbf{AB}) + \dim(N(\mathbf{A})).$$

Solution: You can get the proof of the following statement in the solutions for HW. 2 problem 4 (a):

$$\text{rank}(\mathbf{B}) \leq \text{rank}(\mathbf{AB}) + \dim(N(\mathbf{A})).$$

Now, clearly $N(\mathbf{B}) \subseteq N(\mathbf{B}^\top \mathbf{B})$. Let $\mathbf{x} \in N(\mathbf{B}^\top \mathbf{B})$, then $\mathbf{B}^\top \mathbf{B} \mathbf{x} = \mathbf{0}$, and $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|_2^2 = 0$, which occurs iff $\mathbf{B} \mathbf{x} = \mathbf{0}$, therefore $\mathbf{x} \in N(\mathbf{B})$, and $N(\mathbf{B}^\top \mathbf{B}) \subseteq N(\mathbf{B})$. Thus $N(\mathbf{B}^\top \mathbf{B}) = N(\mathbf{B})$.

Using the rank-nullity theorem, $R(\mathbf{B}^\top \mathbf{B}) = R(\mathbf{B})$, and $\text{rank}(\mathbf{B}^\top \mathbf{B}) = \text{rank}(\mathbf{B})$.